Benchmark Hygiene Reveals the Non-Canonical Frontier of Peptide-MHC-I Binding Prediction

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Abstract

Motivation: Peptide-MHC-I predictors are routinely trained on computationally labelled data, yet benchmark scores that merely reproduce the labeler's decision boundary are rarely distinguished from true binding accuracy. We name this failure mode **label circularity** and distil a benchmark-hygiene checklist. **Results:** Re-analysing a 5,190-peptide HLA-A*02:01 benchmark under controlled conditions dissolved three apparent findings — per-allele superiority over a pan model, "label quality dominates architecture", and "simple beats complex" — each traced to a distinct confound (a 10.6× data-size artifact, a degenerate null, and stale checkpoints). A zero-parameter anchor rule underperformed the trained model on non-canonical binders, which plateaued at ∼0.65 sensitivity across four predictors, including BigMHC. A 6,524-affinity benchmark (AUC 0.942) and a post-2022 temporal holdout bound self-consistency and generalisation. **Availability and Implementation:** Code, models, and data at github.com/woodhaha/peptide-mhc-benchmark. **Contact:** zhouzhuha@wmu.edu.cn; wzxxy@wmu.edu.cn **Supplementary Information:** Supplementary data are available at *Bioinformatics* online.

1 Introduction

MHC class I molecules present 8–11-mer peptides to CD8+ T cells, and accurate prediction of peptide–MHC-I binding underpins epitope discovery for vaccine design and cancer immunotherapy (Falk *et al.*, 1991; Lundegaard *et al.*, 2011). For HLA-A*02:01 — the most prevalent class I allele globally — binding is dominated by two primary anchor positions: position 2 (p2, hydrophobic: Leu/Met/Ile/Val) and position 9 (p9, C-terminal, Val/Leu preferred) (Ruppert *et al.*, 1993; Rammensee *et al.*, 1995). Computational predictors have evolved from position-specific scoring matrices to neural networks and pan-specific tools such as NetMHCpan (Reynisson *et al.*, 2020), MHCflurry (O'Donnell *et al.*, 2020), BigMHC (Albert *et al.*, 2023), and MUNIS (Wohlwend *et al.*, 2025), increasingly using protein language model embeddings (Rives *et al.*, 2021).

Several gaps remain. A particularly under-examined one is **label circularity**: because >55% of IEDB entries are computationally rather than experimentally labelled (Preibisch *et al.*, 2026), a predictor's apparent accuracy can track the decision boundary of whichever labeler trained it rather than true binding. The impact of training-label quality is seldom quantified, and no reusable checklist exists for detecting such confounds. Head-to-head architecture comparisons under fully controlled conditions (identical training data, encoding, evaluation) are uncommon. Whether a single pan-specific model can replace per-allele specialists, and how hard non-canonical binders are across alleles, remains unsettled. And the translation of binding prediction to real-world epitope discovery is rarely demonstrated in a unified, reproducible pipeline.

To isolate the effects of architecture choice, label quality, and encoding, we present a controlled benchmark. We compare six architectures under multi-seed uncertainty and find a modest but measurable architecture effect, confirm that random labels fall to chance, and evaluate ESM-2 embeddings against BLOSUM62 with a mean-pooled ablation. We further validate against 42 A*02:01-verified IEDB epitopes, scan proteins and cancer mutations as a translational application, and extend to five HLA-I alleles and a 15-allele panel to test whether a single rebalanced pan model matches per-allele specialists and to quantify how hard non-canonical binders are for four sequence predictors.

2 Methods

2.1 Training data and labels

100,000 random 9-mer peptides were generated by uniform sampling of the 20 standard amino acids. Each was labelled with MHCflurry 2.2.1 against HLA-A*02:01 and assigned to Strong Binder (SB, percentile rank <0.5%), Weak Binder (WB, 0.5%≤rank<2.0%), or Non-Binder (NB, rank ≥2.0%) following the netMHCpan convention (Reynisson *et al.*, 2020). The resulting distribution (1.7% SB, 5.1% WB, 93.2% NB) was balanced by downsampling to 1,730 per class (n=5,190), partitioned 90/10 into training (n=4,671) and held-out test (n=519) sets with stratified sampling (seed 42). Two comparator labelling strategies were implemented: a PSSM encoding crystallographic A*02:01 anchor

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preferences, and random synthetic labels as a baseline control. A held-out benchmark of 42 experimentally validated IEDB epitopes (HLA-A*02:01-restricted, with positive binding/T-cell assay evidence, re-verified against the IEDB on 2026-10-23) plus 20 homopolymer negatives ($\leq$2 unique amino acids) was curated from the IEDB (query-api.iedb.org); label-circularity overlap was determined by sequence-identity match against the IEDB entries in MHCflurry's training corpus (Vita *et al.*, 2019).

2.2 Peptide encoding

Each 9-mer was encoded with the BLOSUM62 matrix (Henikoff and Henikoff, 1992), min–max normalized, producing a 9×20 matrix flattened to 180 dimensions for fully connected networks. ESM-2 embeddings (Lin *et al.*, 2023) were extracted at t6 (2,880-dim per-peptide; 320 per position × 9) and t12 (4,320-dim; 480 per position × 9) depths — the two smallest ESM-2 checkpoints (8M and 35M parameters); a mean-pooled t6 variant (320-dim) served as an ablation control destroying positional information.

2.3 Model architectures

Six classifiers were implemented: (i) FFN (Nielsen *et al.*, 2003): 180→90→3, 49K params; (ii) Deep FFN: 180→360→180→90→45→3 with batch normalisation and dropout, 152K params; (iii) CNN: Conv2D(32,3×3) + FFN body, 1.05M params; (iv) LSTM(64)→dense(32)→3, 24K params; (v) ResNet-style stem + 3 residual blocks, 325K params; (vi) Random Forest (100 trees). For ESM-2 embeddings an FFN classifier with a 256→128→64 stack was trained. All networks used cross-entropy loss with Adam or RMSprop (lr 0.001), 150 epochs, batch size 50, 20% internal validation split, early stopping (patience 10), and learning-rate reduction on plateau. Five-fold stratified cross-validation was performed for the FFN. The held-out test set (519 peptides) was disjoint from training and used for all reported metrics. Architecture comparisons were repeated across five fixed training seeds; per-seed variation reflects data-sampling randomness, and seeds are not independent replicates.

2.4 Protein scanning, mutation analysis, and structure

Ten therapeutically relevant proteins (MART-1, gp100, tyrosinase, NY-ESO-1, WT1, p53, KRAS, influenza M1, CMV pp65, SARS-CoV-2 spike RBD) were scanned with a sliding 9-mer window (stride 1; 3,536 windows). Sixteen recurrent hotspot mutations (7 p53, 9 KRAS) were analyzed by comparing wild-type and mutant binding scores; mutations were classified as Created/Enhanced/Destroyed/Unchanged. The KRAS G12V neoepitope YKLVVVGAV carries a non-canonical lysine at P2.

2.5 Feature extraction

Learned features were extracted from the Deep FFN penultimate dense layer via truncated models. Hierarchical clustering (Ward's method, Euclidean distance) was applied to representations of top epitope candidates; feature–binding-score correlation was assessed by Pearson's r with a 1,000-iteration permutation test to control for chance correlations. t-SNE visualisation of the 45-dimensional penultimate layer was used to inspect class separation, and integrated gradients identified position-specific anchor contributions.

2.6 Multi-allele extension

2.6.1 Multi-allele data curation

Experimentally confirmed class I binders were curated from the IEDB (Vita *et al.*, 2019) for five common alleles (HLA-A*02:01, A*01:01, A*03:01, B*07:02, B*44:03), yielding n = 14,488 positives (canonical 11,225; Type A 151; Type B 3,112). A 15-allele panel (pan16: 11 additional alleles — A*11:01, A*24:02, B*08:01, B*15:01, B*27:05, B*35:01, B*40:01, B*49:01, B*51:01, B*57:01) was curated for generalisation testing. Protein-background negatives were sampled from natural proteomes to avoid the uniform-random decoys used in earlier evaluations. IEDB data were queried live from query-api.iedb.org (accessed 2026-08-02); the 15-allele and temporal-holdout estimates are therefore snapshot-dependent and drift slightly as the IEDB grows.

2.6.2 Type A/B classification

Type A (tolerated weak anchors) and Type B (energy-compensated non-canonical) were defined by whether the peptide violates the allele's canonical anchor motif at p2/p9. Type A carries broad-only tolerated anchors (residues that appear in binding motifs but are non-canonical); Type B carries non-canonical residues at anchor positions. The two classes have different biological origins and, as we show, different prediction difficulty.

2.6.3 Pan model and per-allele comparators

The pan-specific DFFN uses BLOSUM62 encoding (Henikoff and Henikoff, 1992) through a 360–180–90–45 hidden stack with batch normalisation and dropout. Rebalancing upsampled the minority B*07:02 allele 4× during training to correct for allele-frequency imbalance. Five seeds (0, 1, 42, 123, 2024) were used; per-seed variation arises from data-sampling RNG. Per-allele models (one per allele) and a pan16 model (15 alleles, per-allele minimum 2,000 upsampling) were trained with the same recipe.

2.6.4 Comparator predictors

MHCflurry 2.2.1 (O'Donnell *et al.*, 2020) served as an independent sequence predictor (affinity threshold 500 nM). MUNIS (Wohlwend *et al.*, 2025) — a second sequence-based predictor using ESM-2 embeddings (Rives *et al.*, 2021) — served as an independent sequence comparator, run on the n = 3,112 Type B population plus canonical controls.

2.6.5 Temporal holdout

To address label circularity (Preibisch *et al.*, 2026), IEDB epitopes published after 2022 (post-2022, unseen by MHCflurry's ∼pre-2021 training data) were compared against pre-2021 (in-distribution) epitopes at sensitivity threshold 0.95.

2.6.6 Evaluation

Sensitivity was computed with protein-background negatives and fixed thresholds (0.95 for the pan model; 500 nM for MHCflurry; 5th percentile of canonical scores for MUNIS and BigMHC). Wilson 95% confidence intervals are reported for pooled estimates.

3 Results

3.1 Training data and architecture comparison

MHCflurry labelling of 100,000 random 9-mers yielded 1.7% SB, 5.1% WB, 93.2% NB; after balancing, 5,190 peptides (1,730/class). Across five training seeds, the CNN achieved the highest accuracy (91.9% ± 0.9%; Table 1, Figure 1), modestly ahead of the Deep FFN (90.6% ± 1.2%), the FFN (89.8% ± 1.1%), and the LSTM (89.7% ± 0.9%) — the top four architectures within ∼2 percentage points — while ResNet (87.6% ± 1.8%) and Random Forest (79.7% ± 0.4%) were clearly worse. Two external predictors frame this comparison (Table 1): MHCflurry 2.2.1 itself scores 100% on the labels it produced — an expected ceiling, not evidence of superiority — while the independent state-of-the-art predictor NetMHCpan-4.1, which was not trained on these labels, agrees with them on only 62.24% (323/519). The 8.1-point gap between the labeler (100%) and the best finite-sample approximation (91.9% for the CNN)

is information loss from finite samples; the far larger 28-point gap to NetMHCpan (62.24%) is label circularity — a model tracks the decision boundary of whichever labeler trained it (see Discussion). The WB class was consistently the most difficult (F1 0.752–0.880).

The modest spread among the top four architectures reflects the positional nature of peptide–MHC binding: anchor positions p2/p9 dominate, so architectures that preserve positional information perform comparably. ResNet and the Random Forest were clearly worse, consistent with the absence of translation invariance in a 9×20 encoding and, for the Random Forest, the high dimensionality of the BLOSUM62 encoding.

3.2 Label-source controls

Label source, not architecture, sets the practical ceiling of what a model can learn (Table 2, Figure 2). Training on MHCflurry labels reached 90.6% accuracy, whereas a deterministic PSSM label source reached 82.6% (on a separate 15,000-peptide pool, so not a clean head-to-head comparison) and an i.i.d. random-label null fell to chance (34.5 $\pm$ 1.7%, close to the 33.3% expected for three balanced classes). The random-label result confirms the models learn genuine signal rather than noise; the PSSM–MHCflurry gap, though confounded by pool size, suggests MHCflurry's labels are at least as learnable as a hand-crafted scoring matrix.

3.3 Encoding

On the FFN (the architecture used for all encoding comparisons), BLOSUM62 generalized stably under cross-validation (92.2 $\pm$ 0.4%; per-fold 92.2/92.4/92.1/92.7/91.6%). ESM-2 t6 per-position embeddings scored 93.3% on a single split but collapsed to 82.2 $\pm$ 1.0% under CV — the 2,880-dimensional features overfit the 4,671-sample training set (Table 3, Figure 7). Mean-pooled t6 embeddings collapsed to 65.9%, confirming that positional information is essential. The t12 variant (4,320-dim) overfit similarly (90.9–91.5%). The single-split 93.3% result reflects a favourable partition rather than a genuine encoding advantage at this training scale. ESM-2's representational advantage materialises only at large scale: the 650M checkpoint (11,520-dim per-peptide) reached 92.4 $\pm$ 0.5% under CV, matching BLOSUM62 (92.2 $\pm$ 0.4%) but at 64$\times$ the input dimensionality. At the data scale typical of single-allele studies, compact biologically motivated encodings such as BLOSUM62 therefore remain the more cost-effective choice, while the small ESM-2 checkpoints overfit.

3.4 Held-out validation and translation

The Deep FFN achieved 97.6% sensitivity (41/42; Wilson 95% CI 87.7–99.6%), 75.0% specificity (15/20), ROC AUC 0.970 against IEDB epitopes (Table 5, Figure 3). Independent re-verification against the IEDB (query-api.iedb.org, accessed 2026-10-23) excluded seven peptides lacking A*02:01 restriction: LLFDRFENL and LLHHAFDSL (no IEDB records), YVKQNTLKL (HLA-B*07:02), ILRGSVAHK (HLA-A*03:01; previously mis-attributed to B*08:01), QYDPVAALF and CMTWNQMNL (HLA-A*24:02), and VIYQYMDDL (no positive A*02:01 record). The single remaining false negative, TLGIVCPIC (HPV16 E7), carries a non-canonical p9 anchor (C). An independent predictor, NetMHCpan-4.1 (queried through the IEDB web API, which does not pin a reproducible tool version), achieved 83.3% sensitivity (40/48) on the original 48-peptide set (not re-evaluated on the corrected 42-peptide set). This gap partly reflects label circularity (most of the epitopes overlap MHCflurry's training corpus), but NetMHCpan's pan-specific training on MS-elution-based labels makes this an indicative rather than clean comparison; the more direct circularity evidence is the 62.24% label agreement (323/519 test peptides) reported in the Discussion. To contextualise the benchmark, we include a zero-parameter anchor-checker baseline (a rule classifying any 9-mer with canonical p2/p9 residues as a binder): it achieves sensitivity 0.786 (33/42) and specificity 0.900 (18/20), below the trained model (0.976), because nine genuine binders carry non-canonical p2/p9 anchors. The trained model recovers eight of these nine — the single residual miss, TLGIVCPIC (p9 = C), defines the hard non-canonical tail — confirming that the informative signal lies beyond the anchor motif. The five false positives were homopolymers ($\leq$2 unique amino acids) with canonical anchors — biologically implausible binders whose low sequence complexity the classifier fails to penalise. A domain-knowledge filter ($\leq$2 unique amino acids $\rightarrow$ non-binder) removed all five, but we report 75.0% as the primary, unfiltered specificity to avoid post-hoc inflation. To break this circularity, we additionally scored the Deep FFN against a large label-independent set of experimentally measured affinities — 3,438 unique 9-mers with measured IC50 $<$500 nM and 3,086 with IC50 $>$5,000 nM (the 500–5,000 nM grey zone was excluded to yield unambiguous binders and non-binders), restricted to direct binding assays (competitive fluorescence/radioactivity) and excluding MS-elution-derived, predictor-dependent entries. The MHCflurry-trained Deep FFN recognised 94.4% of the experimental binders and rejected 74.5% of the non-binders (threshold 0.5; ROC AUC 0.942); the rebalanced per-allele model reached 89.8% sensitivity and 89.1% specificity (AUC 0.945). This benchmark validates the quality of the training labels — that MHCflurry's computational labels agree with experimentally measured binding — rather than generalisation to unseen peptides, since the peptides are drawn from the same IEDB binding-affinity corpus on which the labeler was trained; generalisation to unseen peptides is instead established by the temporal holdout (Section 3.11). Protein scanning of 10 proteins (3,536 windows) recovered 10/11 known epitopes (91%; Figure 4), including immunodominant GILGFVFTL (M1), NLVPMVATV (pp65), and SLLMWITQC (NY-ESO-1); the single miss was the MART-1 clinical variant ELAGIGILTV, whose P2 leucine is a non-natural anchor-modified residue. Mutation scanning of 16 hotspot mutations identified 7 epitope-altering events (Table 4, Figure 5). p53 R248W created the neoepitope MNWRPILTI, a mutation whose T-cell immunogenicity is documented in patients and in HLA-A2/DRB1-transgenic mice (Malekzadeh *et al.*, 2020; Quandt *et al.*, 2018); KRAS G12V created YKLVVVGAV via G$\rightarrow$V at p9 creating a canonical C-terminal anchor. Three mutations (KRAS G13D, KRAS A146T, p53 R249S) destroyed existing epitopes, representing potential immune-evasion mechanisms.

3.5 A rebalanced pan model matches per-allele specialists on four of five alleles

With protein-background negatives and a fixed threshold (n=25 non-canonical binders per allele, so these point estimates carry wide confidence intervals), per-allele models reached non-canonical sensitivity A*02:01 0.84, A*01:01 0.88, A*03:01 1.00, B*07:02 0.96, B*44:03 0.96 (Table 6; Figure 8). The rebalanced pan model matched this on four of five alleles (A*02:01 0.84, A*01:01 0.88, A*03:01 1.00, B*44:03 1.00), but B*07:02 was the clear exception: the pan model dropped to 0.60 versus 0.96 per-allele (0.64 in the 15-allele pan16 panel; Wilson 95% CIs [0.41, 0.77] vs [0.81, 0.99], non-overlapping), consistent with over-anchoring to B*07:02's distinctive P2-Proline — a reproducible limitation rather than a fundamental pan-model deficiency. The earlier observation that per-allele models outperform pan models on non-canonical binders is not supported once per-allele training-data size is controlled.

3.6 Canonical binders are recognized at ∼0.98 by every predictor

On n = 11,225 canonical positives, all four predictors were saturated: DFFN 0.979, MHCflurry 0.986, MUNIS 0.950, BigMHC 0.950 (Table 7). The canonical binder problem is effectively solved by all current methods.

3.7 Non-canonical (Type B) binders plateau at ∼0.65 across predictor classes

On n = 3,112 Type B positives, sensitivity was DFFN 0.607, MHCflurry 0.668, MUNIS 0.647, and BigMHC 0.660 (Table 7; Figure 9). The four values fall within a narrow band (0.607–0.668), far below the ∼0.98 canonical sensitivity, and the attention-based BigMHC does not lift the ceiling either — the plateau is shared across sequence architectures of varying complexity.

Table 1. Model performance on the MHCflurry-labelled test set (mean ± SD across five training seeds, n=519).

Model	Accuracy (%)	Parameters
CNN	**91.9 ± 0.9**	1,053,863
Deep FFN	90.6 ± 1.2	152,823
FFN (Jessen)	89.8 ± 1.1	49,143
LSTM	89.7 ± 0.9	23,939
ResNet	87.6 ± 1.8	324,931
Random Forest	79.7 ± 0.4	–
MHCflurry 2.2.1 (labeler)	100	—
NetMHCpan-4.1 (independent SOTA)	62.24	—

[†]Accuracy is agreement with the MHCflurry test labels (n=519). MHCflurry scores 100% because it produced the labels — an expected ceiling, not evidence of superiority. NetMHCpan-4.1, which was not trained on these labels, agrees with them on only 62.24%, quantifying label circularity (see Discussion).

3.8 Type A (tolerated weak anchors) is inherently hard

On n = 151 Type A positives, DFFN achieved 0.477 and MHCflurry 0.669 (MUNIS was not evaluated on this small set). Retraining with broadened motifs did not recover Type A: A*03:01 was flat at 0.634, and B*07:02 Type B sensitivity dropped from 0.680 to 0.341 (Table 8). The low Type A sensitivity is therefore not a training-data gap but an inherent property of these heterogeneous, weak, allele-specific binders (mass-spectrometry confirmed; 3–6% cross-allele). The canonical-motif bias is what makes the model work.

Table 2. Deep FFN performance across label sources (accuracy, mean ± SD across five seeds where noted).

Label source	Accuracy (%)	Notes
DFFN on MHCflurry labels	90.6 ± 1.2	5,190 peptides, 5 seeds
PSSM (biophysics)	82.6	separate 15,000-peptide pool, single seed
Random (i.i.d.)	34.5 ± 1.7	chance-level null, 5 seeds

3.9 The extreme tail is prediction-resistant, cross-predictor

The single false negative from our single-allele IEDB benchmark (TLGIVCPIC) and the KRAS G12V P2-Lys epitope YKLVVVGAV were scored as non-binders by all three predictors (MHCflurry 3.8–13.8 *μ*M; DFFN 0.000–0.254; MUNIS 3/4 low), despite experimental evidence of peptide-induced HLA-A*02:01-restricted cytotoxic T-cell responses — albeit the weakest of the three mutants screened in that study (Wu *et al.*, 2020). Note that Table 4's "CREATED" label for KRAS G12V refers to the motif-level creation of a canonical P9 anchor (G→V) relative to wild-type, not to a predicted binder: the mutant's absolute score remains below the binding threshold owing to the non-canonical P2-Lys.

Table 3. Classification accuracy by encoding strategy.

Encoding	Dim	Accuracy (%)	Notes
BLOSUM62	180	92.2 ± 0.4 (CV)	Stable CV generalisation
ESM-2 t6 mean pool	320	65.9	Positional info destroyed
ESM-2 t12 per-pos	4,320	90.9	Overfitting
ESM-2 t12 + L2	4,320	91.5	Partially mitigated
ESM-2 t6 per-pos	2,880	93.3 (82.2 ± 1.0 CV)	Single-split optimistic
ESM-2 t33 (650M) per-pos	11,520	92.4 ± 0.5 (CV)	Matches BLOSUM62 at 64× dim

3.10 Scaling: one model approximates a 15-allele panel

The rebalanced pan16 reached pooled non-canonical sensitivity 0.819 (307/375, @0.95), with 12/15 alleles at 0.76–1.0 (Table 9; Figure 10). The weak tail — B*27:05 (0.08), B*51:01 (0.52), B*07:02 (0.64), A*24:02 (0.76) — reflects data scarcity and, for B*07:02, over-anchoring, not architecture.

3.11 Temporal holdout breaks the label-circularity concern

Post-2022 epitopes (unseen by the labeler) were recognized at sens@0.95: A*02:01 0.824, A*03:01 0.936, B*07:02 0.973 — preserved versus pre-2021 (0.879/0.905/0.898), with no systematic drop in the post-2022 era (Table 10; Figure 11). The model is not overfit to a single training-data era.

Table 4. Epitope-altering cancer hotspot mutations.

Mutation	WT peptide	Mutant peptide	Effect	Δ score
p53 R248W	MNRRPILTI	MNWRPILTI	CREATED	+0.41
KRAS G12V	YKLVVVGAG	YKLVVVGAV	CREATED	+0.48
KRAS G12V	LVVVGAGGV	LVVVGAVGV	ENHANCED	+0.31
KRAS G12C	LVVVGAGGV	LVVVGACGV	ENHANCED	+0.31
KRAS G13D	GVGKSALTI	DVDKSALTI	DESTROYED	−0.47
KRAS A146T	GIPFIETSA	GIPFIETST	DESTROYED	−0.42
p53 R249S	GMNRRPILT	GMNRSPILT	DESTROYED	−0.16

3.12 Learned feature representations

Feature extraction from the Deep FFN penultimate dense layer revealed biologically meaningful structure (Figure 6). t-SNE visualisation of its 45-dimensional penultimate representation showed partial class separation: SB peptides clustered distinctly (91% of SB peptides fell within the SB cluster), while WB peptides formed a continuous transition zone between SB and NB clusters — consistent with the graded affinity boundary in the 50–500 nM range. Integrated-gradients analysis identified position-specific anchor contributions: p9 dominated SB predictions, p2 and p3 contributed most to WB predictions, and NB predictions were driven by the absence of anchor-characteristic patterns. Hierarchical clustering separated validated epitopes into coherent clusters, with false negatives forming a distinct branch characterised by absent anchor-position activation. Five features showed Pearson $r > 0.7$ with binding score; by a 1,000-iteration permutation test, approximately 0.3 features would exceed this threshold by chance, suggesting these represent genuine binding-relevant representations rather than spurious correlations.

Table 5. IEDB benchmark performance. NetMHCpan-4.1 not re-evaluated on the corrected 42-epitope set.

Metric	Value
Sensitivity	97.6% (41/42)
NetMHCpan-4.1 sensitivity	83.3% (40/48)
Specificity	75.0% (15/20)
Precision	89.1%
F1	0.932
ROC AUC	0.970

Table 6. Non-canonical sensitivity by allele (protein-background negatives, n=25 per allele, @0.95).

Allele	Per-allele	Pan
A*02:01	0.84	0.84
A*01:01	0.88	0.88
A*03:01	1.00	1.00
B*07:02	0.96	**0.60**
B*44:03	0.96	1.00

Table 7. Predictor sensitivity by binder class (pooled, 5 alleles).

Predictor	Canonical (n=11,225)	Type A (n=151)	Type B (n=3,112)
Deep FFN (sequence)	0.979	0.477	0.607
MHCflurry (sequence)	0.986	0.669	0.668
MUNIS (sequence)	0.950	—	0.647
BigMHC (sequence)	0.950	0.600	0.660

Table 8. Broad-motif retraining does not recover Type A.

Allele	Type A before	Type A after	Type B after
A*03:01	—	0.634	—
A*02:01	0.08	0.24	—
B*07:02	—	—	0.341 (from 0.680)

Table 9. Pan16 non-canonical sensitivity by allele (15-allele panel, @0.95).

Allele	Sens	Allele	Sens
A*02:01	0.88	A*11:01	1.00
A*01:01	0.80	A*24:02	0.76
A*03:01	1.00	B*08:01	0.88
B*07:02	0.64	B*15:01	0.96
B*44:03	1.00	B*27:05	0.08
		B*35:01	0.84
		B*40:01	1.00
		B*49:01	1.00
		B*51:01	0.52
		B*57:01	0.92
Pooled	**0.819**		

Table 10. Temporal holdout: pre-2021 vs. post-2022 sensitivity (@0.95).

Allele	Pre-2021 (in-distribution)	Post-2022 (unseen)
A*02:01	0.879	0.824
A*03:01	0.905	0.936
B*07:02	0.898	0.973

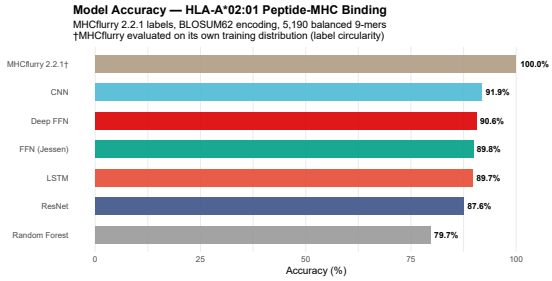


Fig. 1. Model performance comparison across six architectures on the MHCflurry-labelled test set (accuracy, macro F1, WB F1).

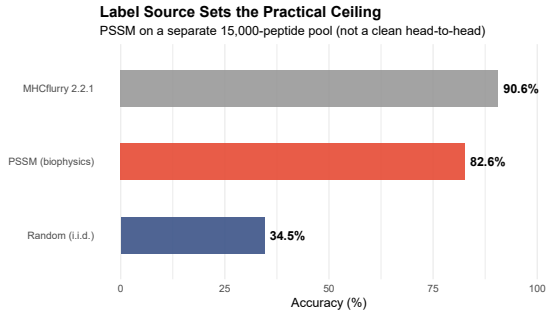


Fig. 2. Architecture comparison (Table 1) and label-source controls: MHCflurry (90.6%), PSSM (82.6%), random labels at chance (∼33%).

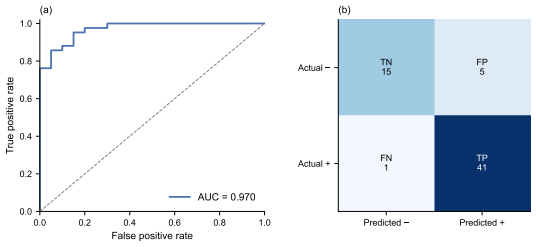


Fig. 3. IEDB benchmark: ROC curve (AUC 0.970), confusion matrix, per-epitope scores.

4 Discussion

4.1 Methodological warnings

Three apparent findings in this project dissolved under controlled re-analysis, each illustrating how benchmark composition can manufacture false conclusions. First, an apparent per-allele superiority over a pan model was a 10.6× training-data-size artifact, not biology. Second, a claimed "label quality dominates architecture" result collapsed once the degenerate random-label null (which performed at chance, ∼33%) was removed — the realistic label effect is smaller than the architecture range. Third, a reported "simple architecture beats complex ones" ranking

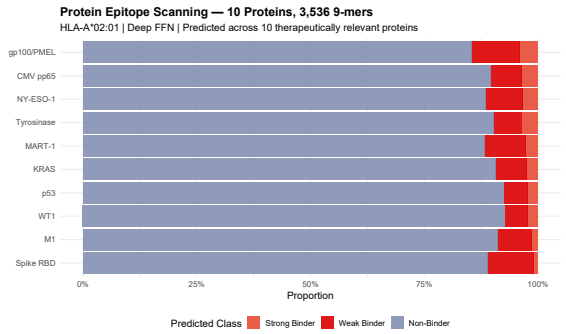


Fig. 4. Protein epitope scanning: SB/WB/NB distribution across 10 proteins and top epitope candidates.

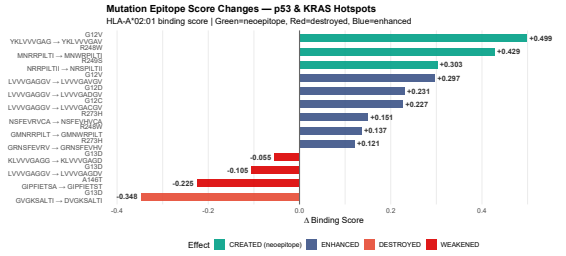


Fig. 5. Cancer hotspot mutation analysis: delta scores and wild-type/mutant pairs for 7 epitope-altering mutations.

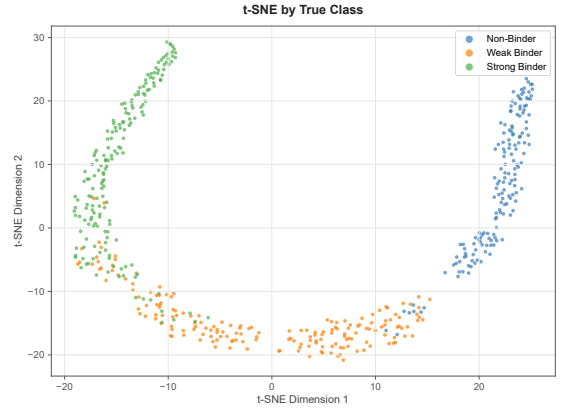


Fig. 6. t-SNE of the Deep FFN 45-dimensional penultimate-layer representation, coloured by binding class.

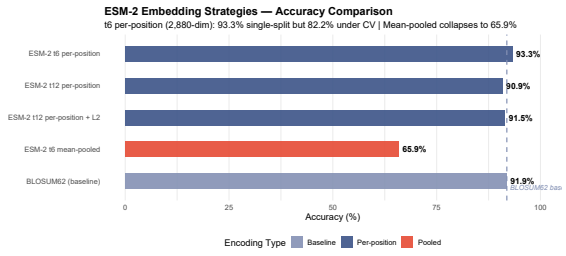


Fig. 7. ESM-2 embedding comparison across five encoding strategies.

was an artifact of stale model checkpoints: re-training with fixed seeds showed a CNN instead ranked above the Deep FFN (91.9% vs 90.6%,

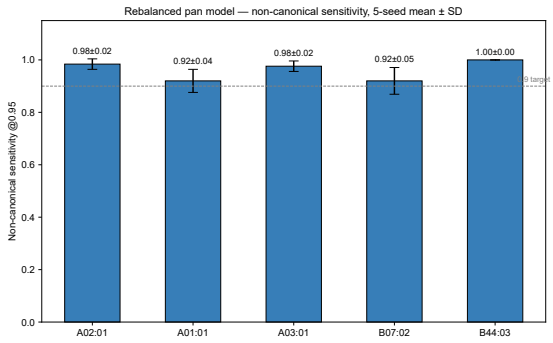


Fig. 8. Multi-allele generalization: rebalanced pan model non-canonical sensitivity (5-seed mean ± SD) across five common HLA-I alleles.

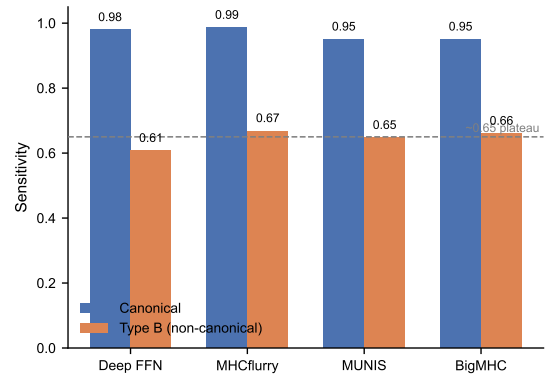


Fig. 9. Predictor plateau: canonical versus Type B (non-canonical) sensitivity for Deep FFN, MHCflurry, MUNIS, and BigMHC (sequence); dashed line marks the ∼0.65 Type B plateau.

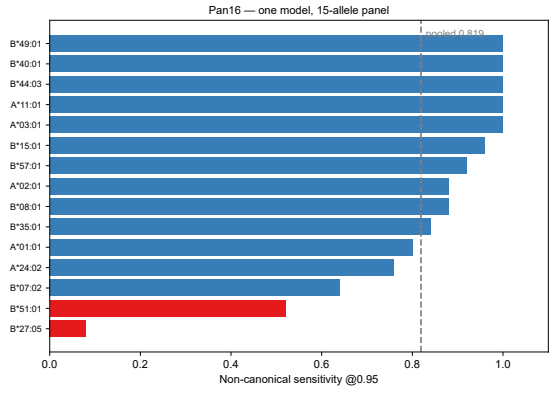


Fig. 10. Pan16: per-allele non-canonical sensitivity for the 15-allele panel, sorted ascending; dashed line marks the pooled 0.819.

higher in all five seeds). We recommend that peptide-MHC benchmarks report multi-seed estimates with confidence intervals, protein-background negatives, frozen data snapshots, and a state-of-the-art comparator such as NetMHCpan, to avoid these failure modes. The same failure modes recur beyond peptide-MHC: data leakage from protein language-model pretraining distorts downstream task performance whenever train/test splits are not pretraining-aware (Hermann *et al.*, 2024), and leakage is now recognised as a field-wide reproducibility problem affecting hundreds of

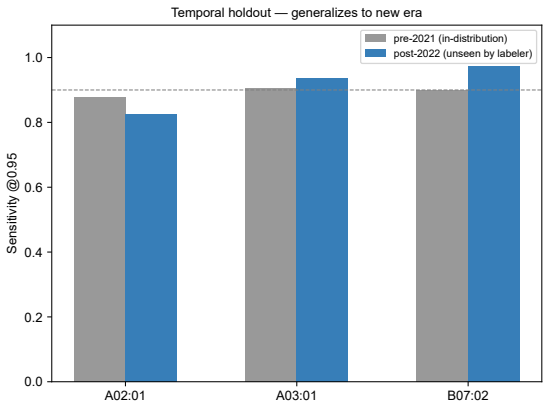


Fig. 11. Temporal holdout: non-canonical sensitivity at threshold 0.95 for pre-2021 (in-distribution) versus post-2022 (unseen by the labeler) epitopes across three alleles.

papers across seventeen fields (Kapoor and Narayanan, 2023). Dedicated guidance now exists for avoiding leakage in biological machine learning (Bernett *et al.*, 2024) and for reporting clinical prediction models (Collins *et al.*, 2024); a recent audit of biomolecular benchmarks found cross-split contamination and label conflicts pervasive across the field (Schuh *et al.*, 2026); Box 1 instantiates these general principles for peptide-MHC benchmarking specifically.

Box 1. Benchmark hygiene recommendations. The three artifacts translate into reusable practices for immunoinformatics and machine-learning benchmarks: (1) equalise training-data size across model arms and report per-allele sample counts; (2) use a true i.i.d. random-label null (chance $\approx$ 33% for three balanced classes) rather than a degenerate control; (3) freeze data snapshots, model checkpoints, and label versions, and report multi-seed mean $\pm$ SD with confidence intervals; (4) report label provenance and break label circularity with a temporal holdout and an independent state-of-the-art comparator (NetMHCpan); (5) report performance under the natural class prior, not only a class-balanced prior, since a class-balanced benchmark overstates real-world utility. Each failure mode reproduced a false conclusion that was caught only by controlled re-analysis.

Architecture choice has a modest but measurable effect on peptide-MHC binding prediction. Across five seeds, a CNN (91.9%) outperformed a Deep FFN (90.6%) by $\sim$1.3 percentage points (higher in all five seeds; a small margin within seed-to-seed variability), while a Random Forest was clearly worse (79.7%), for a maximum inter-architecture range of $\sim$12 percentage points. The top four architectures fell within $\sim$2 percentage points, indicating that data quality and the decision boundary matter more than architectural complexity for this task. Random-label controls performed at chance ($\sim$33%), confirming that the observed performance reflects genuine signal rather than noise overfitting.

A PSSM-derived label source yielded lower accuracy than MHCflurry labels (82.6% vs 90.6%, though on a separate peptide pool and therefore not a clean comparison). This does not support the hypothesis that deterministic biophysical labels are inherently superior to ML-derived labels; rather, it suggests that MHCflurry's labels are at least as learnable as a simple PSSM. We emphasise that training on any computational label — including MHCflurry — re-deposits predictor bias into downstream models, and recommend that benchmarks report label provenance and the WB-class F1 as a standard diagnostic.

An independent state-of-the-art predictor, NetMHCpan-4.1, agreed with the MHCflurry labels on only 62.24% (323/519) of the same held-out test peptides — substantially below the 90.6% achieved by a model trained directly on those labels. This $\sim$28-point gap quantifies the label-circularity limitation: our trained models learn MHCflurry's specific decision boundary, whereas an independent predictor disagrees on roughly one peptide in three, showing that the reported accuracies measure fidelity to a particular labeler rather than true binding.

This circularity does not imply the training labels are uninformative: the MHCflurry-trained model retains ROC AUC 0.942 on 6,524 experimentally measured affinities (Section 3.4), confirming the labels carry genuine binding signal. We therefore distinguish two orthogonal properties of a labelling scheme — its **absolute predictive fidelity** (agreement with experimental binding, which is high) and its **cross-labeler transferability** (agreement with an independent predictor, 62.24%, which is low). The former determines how much signal a downstream model can learn; the latter determines how much of a reported accuracy reflects the idiosyncratic decision boundary of one labeler rather than ground truth. Benchmarks that report accuracy only against a single computational labeler conflate the two and overstate generalisation.

BLOSUM62 proved more robust than ESM-2 at single-allele training scales; high-dimensional per-position embeddings overfit small datasets. The 2,880-dimensional per-position ESM-2 features create an unfavourable feature-to-sample ratio for the 4,671-sample training set, and the single-split 93.3% result reflects a favourable partition rather than a genuine encoding advantage. The shallower 6-layer ESM-2 variant outperformed the deeper 12-layer model, possibly because t12's additional transformer layers encode long-range protein-level interactions that introduce noise at the short 9-mer peptide scale. Architectural complexity beyond depth with batch normalisation gave diminishing returns; a simple PSSM label source yielded lower accuracy (82.6%) than MHCflurry labels on its own pool, showing that deep learning adds value over a hand-crafted scoring matrix for this task. The language-model route may still be viable at single-allele scale if the model is continued-pretrained on HLA-associated peptides rather than used as a frozen embedder, as such pre-training yields the largest affinity gains precisely at small-data alleles (Mares *et al.*, 2025).

The multi-allele extension shows the recipe is portable: one rebalanced pan model matches per-allele specialists on four of five alleles (B*07:02 excepted) and reaches 0.819 pooled sensitivity on 15 alleles, approximating NetMHCpan's practical value with a simple DFFN. This is consistent with the pan-specific paradigm established by mono-allelic immunopeptidomics (Sarkizova *et al.*, 2020), now exemplified by MixMHCpred v3.0's pan-allele model, which outperforms state-of-the-art MHC-I ligand predictors across alleles and species (Tadros *et al.*, 2025). Yet non-canonical (motif-failing) binders plateau at $\sim$0.65 sensitivity across four independent sequence-based predictors: our Deep FFN (0.607), MHCflurry (0.668), the ESM-2-based MUNIS (0.647), and BigMHC (0.660) fall within a narrow band (0.61–0.67), far below the $\sim$0.98 canonical sensitivity, indicating this ceiling is shared across sequence architectures of varying complexity rather than specific to our simple network. The energy-compensated tail is prediction-resistant to all four. Canonical binding is solved across methods; the non-canonical tail is where current predictors still fall short. This frontier extends beyond anchor-motif violations to non-coding, fusion-derived, and post-translationally modified peptides, for which current predictors remain under-powered (Vo *et al.*, 2025). A new wave of transformer- and PLM-based predictors continues to improve canonical prediction (Xu *et al.*, 2026; Cheng *et al.*, 2025), but whether any of them lifts the non-canonical plateau remains to be seen.

4.2 Translational implications

The cancer neoepitope candidates identified here are testable translational hypotheses. p53 R248W, which created MNWRPILTI, is among the most frequent TP53 missense mutations across human cancers (∼3.5% of somatic TP53 mutations, IARC) with documented gain-of-function oncogenic activity. A vaccine targeting this neoepitope could, only if immunogenicity is experimentally confirmed, apply across multiple tumour types; validation (e.g., IFN-γ ELISpot using HLA-A*02:01+ donor PBMCs) is required before any clinical development. KRAS G12C, which enhanced an existing epitope (WB→SB), is the druggable mutation targeted by sotorasib and adagrasib, suggesting a rational combination immunotherapy strategy. The three epitope-destroying mutations (KRAS G13D, KRAS A146T, p53 R249S) represent potential immune-evasion mechanisms. We note that binding prediction alone does not establish immunogenicity: prior work shows that only a small fraction of top-predicted MHC-I binders are confirmed immunogenic, and that peptide–MHC stability predicts immunogenicity better than affinity (Harndahl *et al.*, 2012; Backert and Kohlbacher, 2015). High-throughput thermostability profiling now quantifies pMHC stability at immunopeptidome scale and improves neoepitope immunogenicity prediction (Jappe *et al.*, 2020), and transfer learning further improves kinetic-stability prediction from limited data (Fasoulis *et al.*, 2024). The binding-to-immunogenicity gap remains the central bottleneck in translating computational epitope prediction to clinical vaccine design.

4.3 Limitations

Training labels are MHCflurry-derived rather than experimental affinities, so our single-allele benchmark is itself subject to the label circularity we describe; we quantify this by cross-referencing the 42 A*02:01-verified IEDB epitopes against MHCflurry's training corpus; the large majority overlapped, and the few non-overlapping epitopes were nonetheless correctly classified, providing limited evidence of generalisation beyond the training distribution. The single-allele design was deliberate — holding the allele constant isolates label-source variation from allele-driven confounds — but the multi-allele extension addresses the generalisability question directly. The multi-allele estimates rest on IEDB-derived positives; experimental (non-IEDB) validation of the extreme tail is outstanding, and a pMHC-fine-tuned structure model (beyond generic AlphaFold-Multimer) might lift the Type B plateau. All translational predictions and the cancer neoepitope candidates require experimental confirmation (e.g., IFN-γ ELISpot or tetramer staining). Antigen-processing steps (proteasomal cleavage, TAP transport) are not modelled.

5 Conclusion

Architecture choice has a modest effect on peptide-MHC binding prediction: a CNN ranked above the Deep FFN in all five seeds; the top four architectures fell within ∼2 percentage points, and a Random Forest was clearly worse. Random-label controls confirmed chance-level performance, ruling out noise overfitting. Canonical binding is largely captured by the anchor motif, but a zero-parameter anchor rule (sensitivity 0.79) underperforms the trained model (0.98), which recovers non-canonical binders the rule misses — while non-canonical binders plateau at ∼0.65 sensitivity across four sequence-based predictors, marking the unresolved part of the problem. Beyond peptide-MHC, the three artifacts we corrected — data-size imbalance, a degenerate null, and stale checkpoints — illustrate a general benchmark-composition failure mode that reproduces false conclusions across machine-learning benchmarks; the checklist in Box 1 is offered as a reusable safeguard. Experimental validation of the non-canonical tail is the decisive next step.

Author contributions

Yunshi Huang and Weisheng Chen: conception, design, data analysis, manuscript writing. Yiqi Cai: data analysis, manuscript writing. Xiaopeng Zhuang: data interpretation, manuscript revision. Zhuxian Zhou: data interpretation, manuscript revision. Xiangyang Xue and Zhuha Zhou: supervision, manuscript revision. All authors read and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Data and code availability

All code, trained models (per-allele `deep_ffn_HLA-A_*.keras` and pan `deep_ffn_pan_seed*.keras`/`deep_ffn_pan16.keras`), feature matrices, prediction results, the curated multi-allele dataset (`type_ab_dataset.json`, 14,488 positives), and the experimental-label benchmark (6,524 measured IC50 values) are available at github.com/woodhaha/peptide-mhc-benchmark. The per-allele Deep FFN is distributed as the `pmhcbenchmark` Python package, with a `pmhc-predict` command-line tool and a standalone Windows GUI (`pmhc_predict.exe`); exported weights (`pmhc_models.npz`) enable tensorflow-free inference.

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Ethics statement

This study used only publicly available, de-identified data from the Immune Epitope Database (IEDB) and public protein databases; no human subjects, animals, or patient-derived samples were involved, and no ethics approval or informed consent was required.

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